

Lab: The Cardboard Grabber

Goal

To build a mechanical Actuator (Gripper) that can pick things up.

What You Need

- Cardboard strips.
- Brass fasteners (or string/tape).
- Scissors.

Steps

1. The Linkage (Scissors Mechanism)

Cut two long strips of cardboard. Poke a hole in the middle of both. Connect them with a fastener so they form an "X".

2. The Actuation

Hold the bottom two handles of the "X". Squeeze them together. Watch the top two ends close! Open them... the top opens! You made a **Lever**.

3. The Test

Put a crumpled paper ball on the table. Use your Grabber to pick it up. Can you move it to a cup?

My Grabber Log

Item	Did I pick it up?
Paper Ball	Yes
Pencil	Yes (Tricky!)
Heavy Rock	No (Too heavy)

What We Learned

Robots use simple machines (Levers, Gears) to turn motion into action. Your hand is a machine too!